L16 HW

- Due Mar 7 at 11:59pm
- Points 10
- Questions 8
- Time Limit None
- Allowed Attempts 3

Instructions

You get three attempts on all homework quizzes.

In all quizzes and homeworks in this course, round your answers to THREE DECIMAL places unless otherwise indicated.

Take the Quiz Again

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	13 minutes	10 out of 10

Answers will be shown after your last attempt

Score for this attempt: 10 out of 10
Submitted Mar 6 at 12:50pm
This attempt took 13 minutes.

Question 1
1 / 1 pts

Open the file **NailStrength.xlsx (https://byui.instructure.com/courses/398828/files/159316684/download?wrap=1) (https://byui.instructure.com/courses/398828/files/159316684/download?download_frd=1)**

The file contains data comparing the withdrawal strength of three types of nails. Which pair of nails seem to have different strengths? Use the Bonferroni method and mark all that apply below.

- The Annularly threaded and the Helically threaded.

- ○ The Annularly threaded and the smooth threaded.
- ○ The smooth threaded and the Helically threaded.
- ◉ None are different. All the means seem to be the same.

A study is made of the effect of curing temperature on the compressive strength of a certain type of concrete. Five concrete specimens are cured at each of four temperatures, and the compressive strength of each specimen is measured (in MPa).

The data is found in the file **Temperature_Concrete.xlsx (https://byui.instructure.com/courses/398828/files/159316686/download?wrap=1)** **(https://byui.instructure.com/courses/398828/files/159316686/download?download_frd=1)**

Use this information for all the parts.

Question 2

1 / 1 pts

Part 1: In order to check the requirements for this ANOVA test, what do I have to check? **(Mark all that apply)**

- [ ] That the means are equal
- [x] That the population variances are equal
- [ ] That the distribution of each group is an F distribution
- [x] That the distribution of each group is a normal distribution
- [x] That the data was a simple random sample.

Question 3

1 / 1 pts

Part 2: Regardless of whether or not the requirements are met, run ANOVA to test if the 4 temperature means are the same. What is your test statistic?

59.731

Question 4

1 / 1 pts

Part 3: What is your conclusion at the 0.05 significance level?

- ○ There is enough evidence to show that mean strength for all temperatures are the same.
- ○ There is enough evidence to show that mean strength for all temperatures are different from each other.
- ◉ There is enough evidence to show that at least one of the temperature mean strength is different than the others.
- ○ There isn't enough evidence to show that at least one of the temperature mean strength is different than the others.

Question 5

2 / 2 pts

Part 4: If your conclusion above was to reject the null, and you know there may be a difference in the means, which of the temperature means seem different than each other? Use the Bonferroni method to check. **(Mark all that apply)**

- [ ] 0 and 10
- [x] 0 and 20
- [x] 0 and 30
- [x] 10 and 20
- [x] 10 and 30
- [ ] 20 and 30
- [ ] None are different. They are all the same.

Question 6

2 / 2 pts

In the **CollectorPower.xlsx (https://byui.instructure.com/courses/398828/files/159316688/download?wrap=1) (https://byui.instructure.com/courses/398828/files/159316688/download?download_frd=1)** data you worked on in the previous homework, you rejected the null hypothesis and concluded that at least one of the means was different. Use the Bonferroni method and determine which of the means seem different? (Mark all that apply)

- [ ] A and B
- [x] A and C
- [x] A and D
- [x] B and C
- [x] B and D
- [ ] C and D
- [ ] None are different, they are all the same.

Question 7

1 / 1 pts

Suppose you had 5 groups to compare the means of. ANOVA gave you a small p-value, so you want to do pairwise comparisons between the mean. How many pairs of means would you need to compare?

10

Question 8

1 / 1 pts

Which of the following statements is/are correct? **(Mark all that apply)**

- [x] We only run Bonferroni if we know there may be a difference in the treatment means.
- [ ] Doing pairwise comparisons by doing multiple independent sample t-test is the same as doing Bonferroni.
- [x] Doing pairwise comparisons by doing multiple independent sample t-test is not good because it makes our type I error big.
- [ ] If we had a treatment with 3 levels, I would need to do 3! or 6 different comparisons.
- [x] If we had a treatment with 4 levels, I would need to do 6 different comparisons.

Quiz Score: 10 out of 10

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